

Semantic Data Management for AI

Project 2

Advanced Databases (BDA-GIA)

Context



Data Science Projects

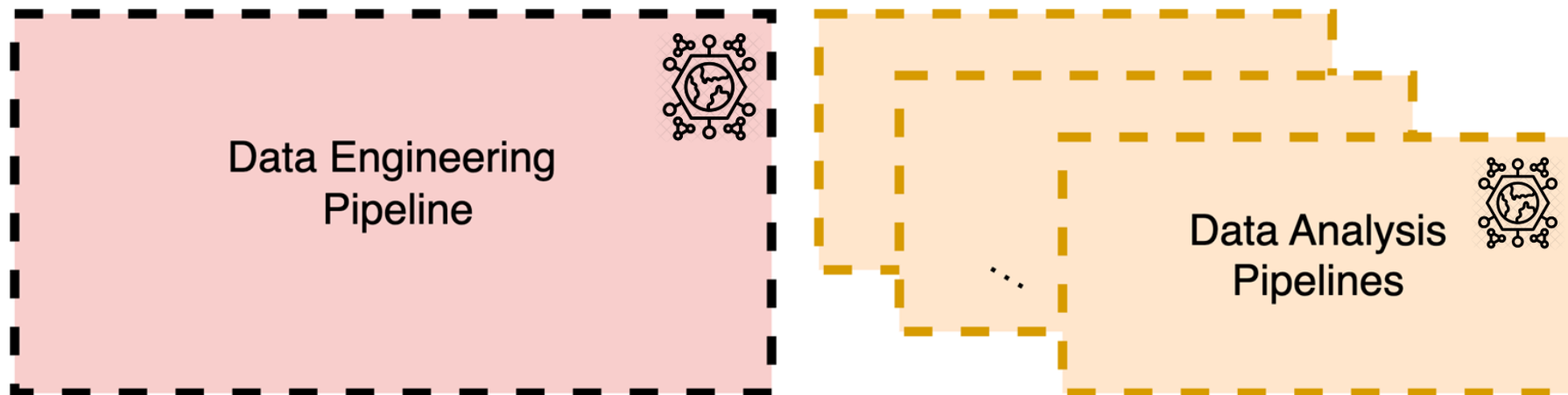
Data science projects require creating **systems** that deploy **data pipelines** spanning three different areas:

- **Business understanding (domain)**
 - What do we want to analyse?
 - What is the added value of an analytical question for the organisation?
- **Data management**
 - Data discovery
 - Data modeling
 - Data storage
 - Data processing
 - Data querying
- **Data analysis**
 - Data preparation
 - Modeling
 - Validation
 - Explainability
 - Visualization

Operationalizing Data Science Pipelines

A common Data Engineering pipeline (DataOps)

- The ad-hoc pipelines do not scale
 - In terms of users (data scientists)
 - In terms of data (volume)
 - In terms of semantics (variety)
- A systematic approach is required to prepare data for data scientists
 - Enriching the heterogeneous data with semantics



DataOps in a Nutshell

The data engineering pipeline (common for the whole organisation)

- Ingests and stores external data into the System
 - Data collection
 - Data Integration (standardization and data crossing)
 - Syntactic homogenization of data
 - Semantic homogenization of data
 - Data quality
 - Clean data / eliminate duplicates
- Expose a cleaned and centralized repository

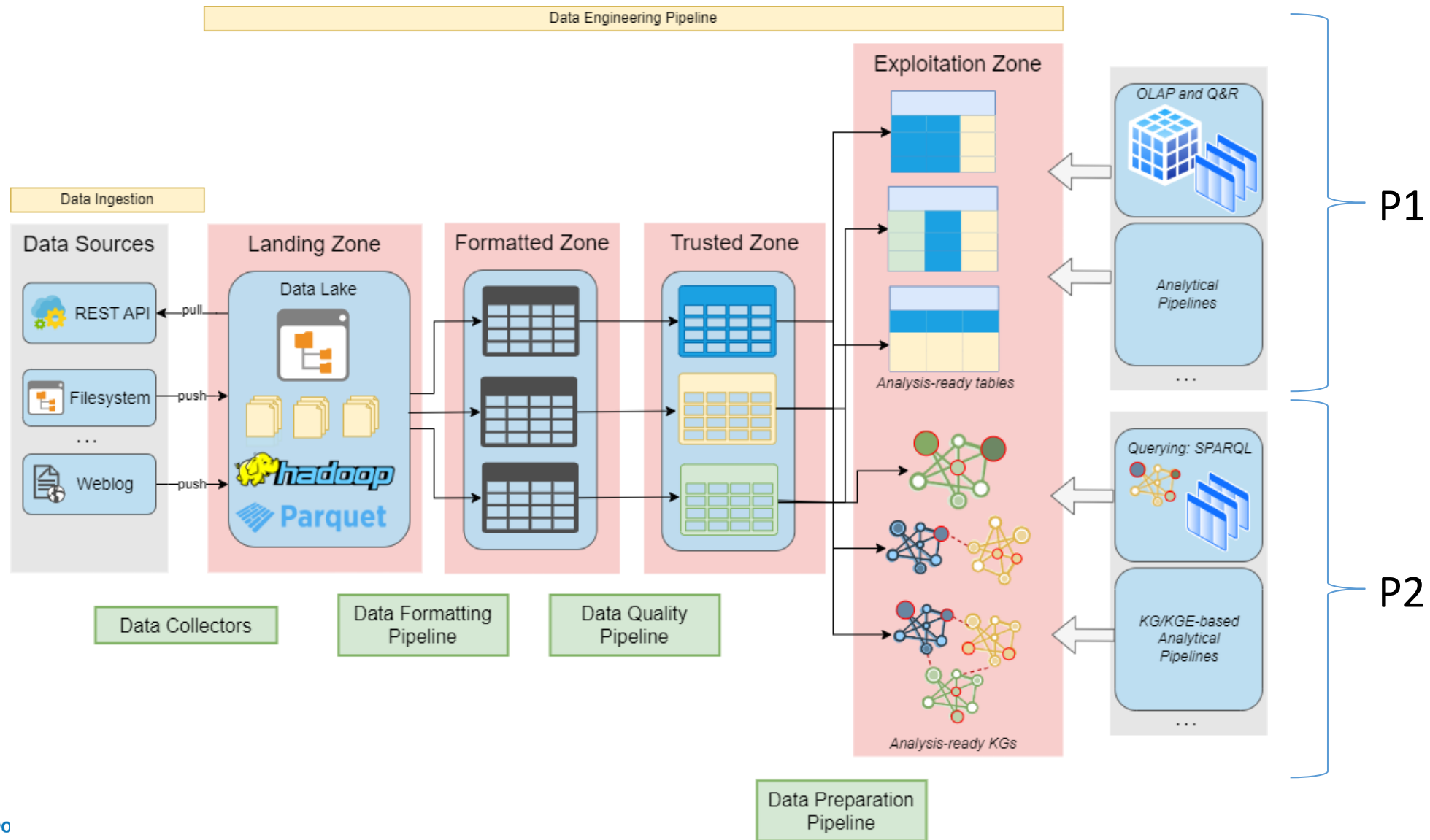
The data analysis pipelines (one for every analytical need)

- Extract a data view from the centralized repository
- Feature engineering
- Specific pre-processing for the given analytical task at hand
 - Labeling
 - Data preparation specific for the algorithm chosen
- Create test and validation datasets
- Learn models (either descriptive statistical analysis or advanced predictive models)
- Validate and interpret the model

The Data Engineering Pipeline

With Knowledge Graphs (KGs)

The Data Engineering Pipeline (s)



The Data Engineering Pipeline

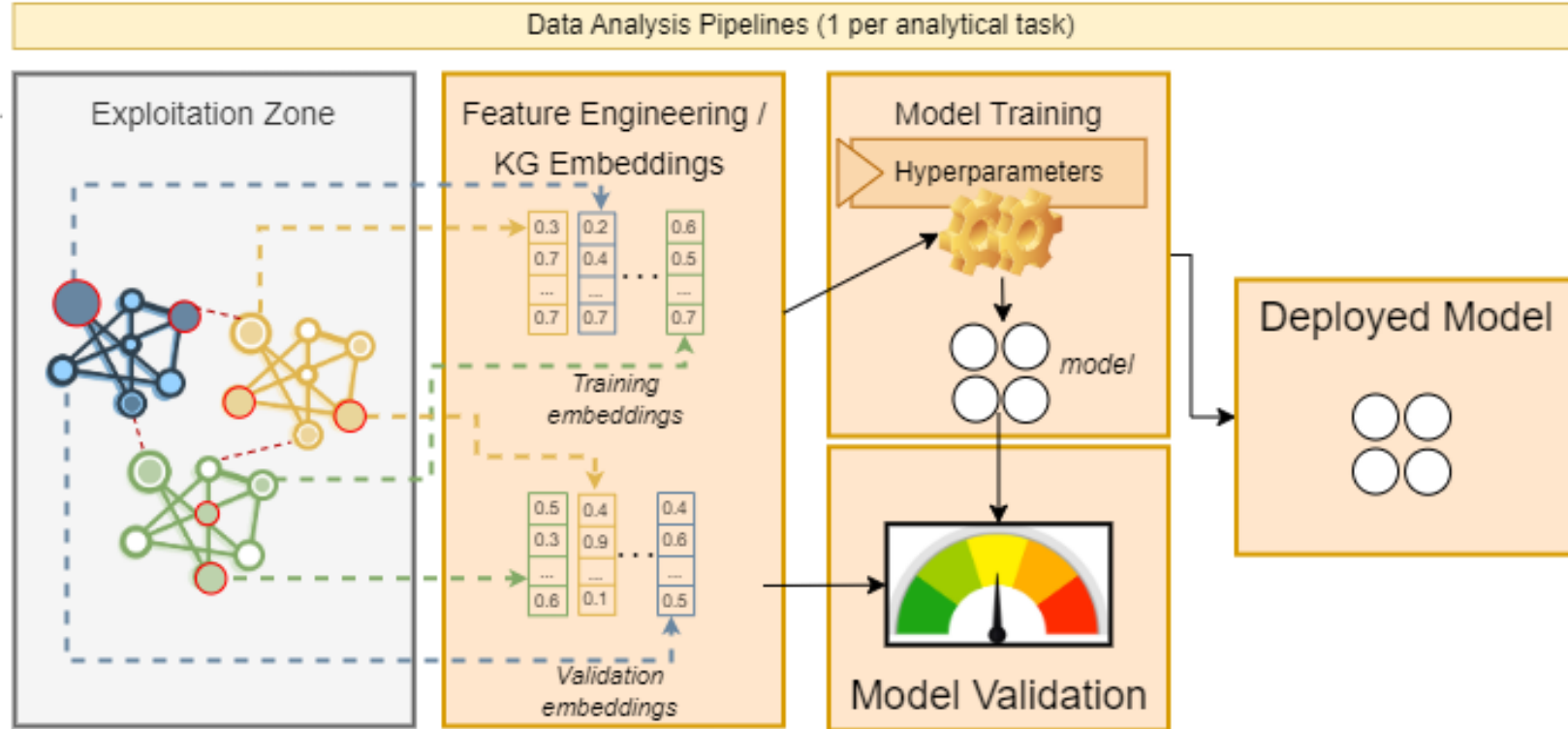
- **Landing Zone – Trusted Zone**, as in P1,
 - ... but KGs could be pushed towards the formatted zone as well.
 - In P2, we will assume **Landing Zone – Trusted Zone** is already done (if possible...)
- The **Exploitation Zone** exposes data ready to be consumed / analysed either by advanced analytical pipelines or external tools.
 - This zone involves semantic homogenization:
 - **Knowledge graphs** can boost semantic homogenization by establishing a common semantics (common vocabulary) for all the data coming from different sources.
 - First step is to **model and load** KGs from previously formatted and cleaned data by combining the instances from the Trusted Zone.
 - Importantly, these KG should benefit from a common vocabulary to further boost data integration.

The Data Analysis Pipelines

With Knowledge Graphs (KGs)

The Data Analysis Pipelines

Generated in the data
engineering pipeline



The Data Analysis Backbone

Unlike the data management backbone, there is an analytical pipeline per analysis need. Thus, a project may define several analytical pipelines

- In the case of KGs, **feature generation** process will include the generation of **KG embeddings** (**vector representation of graphs**), which will further serve as features for applying ML tasks.
 - As result, two corpus of datasets are generated: the training and validation datasets
- **Model training** as before requires choosing an algorithm and specifying the required algorithm hyperparameters, and this outputs a model
- Then, the generated model is **validated** according to some quality criteria (e.g., accuracy, recall, etc.)
- Out of the models generated, one is chosen to be deployed

Objectives of the project

Implementation of an end-to-end Data Science pipeline with Knowledge Graphs

Objectives

- The objective of the project is to boost your previously created data science pipelines with Knowledge Graphs
 - A Data Engineering pipeline should be updated so that the Exploitation Zone is built using KGs.
 - Data Analysis pipelines should focus on exploiting data modelled as KGs
- You choose the analytical tasks
 - Be ambitious!
 - Optionally, if your previous models were based on relationships, try to compare them with KG-based model.
- Some candidate data sources
 - Open Data Portals: OpenDataBCN, Dades Obertes L'Hospitalet, ...
 - Third party APIs: Google Maps, TripAdvisor, Twitter, ...
 - Publicly available: Kaggle, Data.World, AWS Data Marketplace, ...

The Landing Zone

- Implement a **Data Collector** for each data source
 - Download a dataset
 - Convert it to a suitable format and give it a meaningful name
 - Importantly: the Data Collector should allow for periodic executions
- Decide on the organization of your Data Lake
 - No need to use HDFS, you can use Google Drive or your Local File System
- Several datasets might be ingested from one data source

Reuse P1
(if possible)

The Formatted Zone

- Data in the Data Lake must be homogeneized into a common data format
 - The Relational Data Model
- You choose the technology and the design of the database
 - PostgreSQL
 - DuckDB
 - Delta Lake
 - ...
- The Data Formatting Pipeline must be implemented using Spark / SparkSQL (either the SQL or the DataFrame API)
- As a rule of thumb, there should be one table per dataset

Reuse P1
(if possible)

The Trusted Zone

- Three main tasks are conducted here
 - Identification of Data Quality rules on your datasets
 - An assessment of the Quality of the Data
 - Application of Data Cleaning processes (individually per dataset)
- Data quality rules can be expressed as Denial Constraints (explained in class)
- The Trusted Zone will store the same tables as the Formatted Zone where their quality has been improved
- The Data Quality Pipeline must be implemented using Spark / SparkSQL.

Reuse P1
(if possible)

The Exploitation Zone (KG-based)

- Here we will model data as KGs, load them from different datasets in the **Trusted Zone** and integrate them so they are ready to be consumed by the **Data Analysis Pipelines**.



The Exploitation Zone (KG-based)

- Tasks (The focus should be on modelling the input data as KGs):
 - Select the appropriate KG language to do so (RDF/RDFS)
 - Design graph schema with a meta-model definition
 - Create pipelines to populate the graph(s) from the Trusted zone
- Technologies
 - GraphDB
 - RDFLib / Jena
 - ...

The Data Analysis pipelines

- Implement at least two Data Analysis pipelines
 - 1 based on pattern matching queries using **SPARQL**
 - 1 based on running an ML algorithm over the **KG embeddings**
- It can be any query/ML algorithm of your choice, focusing specifically on those where data modelled as graph can make a difference!
- Technologies
 - GraphDB (Querying)
 - PyTorch (Graph embeddings, ML)
 - Scikit Learn (ML)
 - ...

Organization

Teams

- Groups of 3 students
 - Create teams in LearnSQL

Timeline

- April 24th
 - Project statement
- May 7th
 - Follow-up: Exploitation zone: KG modelling – RDF
- May 15th
 - Follow-up: Exploitation zone: KG modelling – RDFS
- May 29th
 - Follow-up: Data Analysis pipelines: SPARQL and KG embeddings
- June 3rd
 - Final submission
- These dates are indicative of the expected progress. Follow-up sessions are not mandatory to attend

What to submit?

- All the code required to deploy your Data Science project
 - A single notebook
 - A set of notebooks
 - A set of Python scripts
- All design choices and non-obvious decisions **must be explained and documented**
 - In the notebooks themselves
 - As a companion document (max 5 pages)

Closing